

HealthConnect Clinic – Week 6

Advanced Analytics & Decision Support Report

Executive Summary

Week 6 builds directly on the Week 5 HealthConnect Clinic dashboard. The existing descriptive analysis was validated and extended through deeper investigation of booking lead time and its relationship with previous no-show history, appointment type, distance and reminder channel. No-show rates increased across lead-time bands, reaching 60.5% for appointments booked 30+ days ahead. The most specific segment identified was 30+ days combined with previous no-show history, with a 67.9% observed no-show rate. Findings were translated into decision support and shared with Data Science as a candidate interaction feature for Week 7 testing.

Week 5 → Week 6 Transition

Week 5 established the baseline KPIs and highlighted booking lead time and previous no-show history. Week 6 continued that work rather than restarting the project, adding relationship-based segmentation across previous no-shows, reminder channel, appointment type and distance.

Objective and Analytical Question

Objective: move from descriptive reporting toward deeper analytical insight and practical decision support. Analytical question: How does booking lead time relate to other factors associated with higher no-show rates?

Validated KPI Baseline

- Overall No-Show Rate: 48.5%
- Repeat No-Show Rate: 55.4%
- Lost Slot Rate: 53.7%
- Highest Lead Time 30+ Days: 60.5%

Advanced Analysis and Findings

Booking lead time: 0–3 days = 24.8%; 4–14 days = 32.6%; 15–30 days = 43.2%; 30+ days = 60.5%. Within 30+ days, previous no-show = 67.9% versus no previous no-show = 55.2%. Within 30+ days, Follow-up = 65.1%, Diagnostic Test = 59.6%, Specialist Consultation = 59.2%, General Consultation = 58.1%. For distance, 15km+ = 66.4%, Unknown = 64.3% (42 appointments; interpret cautiously), <5km = 61.4%, and 5–15km = 58.2%. For reminders, Email = 61.9%, WhatsApp = 60.2%, SMS = 58.0%. Reminder differences are descriptive and do not establish causation.

Key Week 6 Insights

- Booking lead time is the strongest observed pattern, increasing from 24.8% for 0–3 days to 60.5% for 30+ days.
- Previous no-show history strengthens the long-lead pattern: 67.9% versus 55.2% within 30+ days.
- Long lead time remains elevated even without previous no-shows: 55.2%, 6.7 percentage points above the 48.5% baseline.
- Follow-up appointments and the 15km+ group are additional segments for attention within the 30+ day group.
- The combined long-lead + previous-no-show segment created a specific analytical input for Data Science.

Observed No-Show Risk Segments

For decision support, the dashboard uses observed no-show rates to identify segments that may warrant attention. The full comparison table is retained so higher- and lower-observed segments can be viewed against the 48.5% overall baseline.

Decision Support and Recommendations

- Introduce earlier confirmation for appointments booked more than 30 days in advance.
- Prioritise repeat no-show patients within the long-lead segment.
- Give additional attention to long-lead Follow-up appointments.
- Investigate travel-related barriers for long-lead appointments, particularly for patients 15km+ from the clinic.
- Test reminder strategies rather than assuming one channel is more effective.
- Collaborate with Data Science to test the Lead Time × Previous No-Show interaction in Week 7.

Dashboard Enhancement

The Week 5 Power BI dashboard was enhanced rather than replaced. The Week 6 page, “HEALTHCONNECT | ADVANCED NO-SHOW ANALYSIS,” brings together the overall no-show baseline, 30+ day no-show rate, repeat no-show rate, lost slot rate, and relationship views for lead time against previous no-show history, reminder channel, appointment type and distance.

Limitations

The analysis is primarily descriptive and does not prove causation. Reminder-channel differences may reflect non-random assignment. Small segments require caution. The analysis identifies candidate predictive signals but does not validate a predictive model. The proposed interaction feature must be tested before adoption.

Week 7 Testing Requirements

- Test an explicit interaction between extended booking lead time and previous no-show history.
- Compare model performance before and after adding the interaction.
- Evaluate whether the feature improves predictive performance.
- Continue segment-level validation and dashboard refinement.
- Use results to refine decision-support recommendations.

Conclusion

Week 6 strengthened the HealthConnect Clinic solution by moving from individual descriptive patterns toward relationship-based analysis and decision support. The clearest finding was the 67.9% observed no-show rate among 30+ day appointments for patients with previous no-show history. This provides targeted operational insight and a concrete Data Science input for Week 7.